

3.1

$$3.10) ① \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\} \subseteq A$$

$$② \forall x \in A, \{0x, 1x, 2x, 3x, 4x, 5x, 6x, 7x, 8x, 9x\} \subseteq A$$

$$(12) ① \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\} \subseteq A$$

$$② \forall x \in A \text{ 且 } x \neq 0, \{x0, x1, x2, x3, x4, x5, x6, x7, x8, x9\} \subseteq A$$

$$(13) ① 0, 10 \in A$$

$$② \forall x0 \in A, x10, x00 \in A \text{ (例数第2位加1或0)}$$

$$(14) ① 5 \in A$$

$$② \forall x \in A, x+5 \in A$$

4. 真: 2, 3, 4.

假: 1, 5, 6, 7

5. 真: 1, 2, 4, 6, 10, 11, 12, 14

其余为假

6. 可以. $A = \emptyset, B = \{\emptyset\}$ 7. 不能: $A \subseteq B, B \in A$ $\therefore B \in B$, 不存在这样的B集合.

3.2

$$3.11) \text{ 不一定: } A = \{1, 2, 3\} \quad B = \{2\} \quad C = \{3\}$$

$$\text{② 不一定: } A = \{1\}, B = \{1, 2\} \quad C = \{1, 3\}$$

(3) 一定, 证明如下:

$$\text{由 } A \oplus B = A \oplus C,$$

$$\text{有 } A \oplus A \oplus B = A \oplus A \oplus C$$

$$\text{又 } A \oplus A = \emptyset$$

$$\therefore B = C.$$

$$4. 1) \cup A = \{a, b, c\}$$

$$2) \cap A = \emptyset$$

$$3) \cap \{ \cup A \} = \cap \{ \{a, b, c\} \} = \{a, b, c\}$$

$$4) \cup \{ \cap A \} = \cup \{ \emptyset \} = \emptyset$$

$$5. 1) P(A) = \{ \emptyset, \{ \emptyset \} \}$$

$$2) P(A) = \{ \emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\} \}$$

$$3) P(A) = \{ \emptyset, \{a, b\}, \{c\}, \{ \{a, b\}, \{c\} \} \}$$

$$4) P(A) = \{ \emptyset, \{ \emptyset \}, \{ \{ \emptyset \} \}, \{ \{ \{ \emptyset \} \} \}, \{ \emptyset, \{ \emptyset \} \}, \{ \{ \emptyset \}, \{ \{ \emptyset \} \} \}, \{ \emptyset, \{ \{ \emptyset \} \} \}, \{ \{ \emptyset \}, \{ \emptyset, \{ \emptyset \} \} \} \}$$

$$5) P(A) = \{ \emptyset, \{a\}, \{b\}, \{a, b\} \}$$

3.3

$$1. 1) A \cap (B - A) = A \cap (B \cap \bar{A}) = (A \cap \bar{A}) \cap B = \emptyset$$

$$(3) A - (B \cup C) = A \cap \overline{B \cup C} = A \cap (\bar{B} \cap \bar{C})$$

$$= (A \cap \bar{B}) \cap (A \cap \bar{C}) = (A - B) \cap (A - C)$$

$$(5) (A - B) - C = (A \cap \bar{B}) \cap \bar{C} = A \cap (\bar{B} \cap \bar{C})$$

$$= A \cap \overline{B \cup C} = A - B \cup C$$

$$5. 1) A \subseteq \overline{B \cap C}. \text{ 证:}$$

$$\text{左边} = A \cap \bar{B} \cup A \cap \bar{C} = A \cap (\bar{B} \cup \bar{C}) = A \cap \overline{B \cap C} = A$$

$$\therefore A \subseteq \overline{B \cap C}$$

$$(3) A \subseteq \overline{B \cup C}. \text{ 证:}$$

$$\text{左边} = A \cap \bar{B} \cap \bar{C} = A \cap (\bar{B} \cap \bar{C}) = A \cap \overline{B \cup C} = A$$

$$\therefore A \subseteq \overline{B \cup C}$$

$$(5) A \subseteq B \oplus C. \text{ 证:}$$

$$\text{左边} = (A \cap \bar{B}) \oplus (A \cap \bar{C}) = (A \cap \bar{B}) \cup (A \cap \bar{C}) - A \cap \bar{B} \cap \bar{C}$$

$$= A \cap (\bar{B} \cup \bar{C}) - A \cap (\bar{B} \cap \bar{C}) = A \cap (\bar{B} \cup \bar{C} - \bar{B} \cap \bar{C})$$

$$= A \cap (\bar{B} \oplus \bar{C}) = A \cap (B \oplus C) = A$$

$$\therefore A \subseteq B \oplus C$$

$$(7) A = B.$$

$$\text{充分性: 由 } A = B, A \cap B = A, A \cup B = A, \therefore A \cap B = A \cup B$$

$$\text{必要性: } \because A \cap B = A \cup B$$

$$\therefore \begin{cases} A \cap B \cap A = (A \cup B) \cap A & \text{即 } A \cap B = A \\ A \cap B \cap B = (A \cup B) \cap B & \text{即 } A \cap B = B. \end{cases}$$

$$\therefore A = B.$$

4

$$(9) A=B$$

$$\text{必要性: } A-B=B-A. \therefore A \cap \bar{B} = B \cap \bar{A}$$

$$\therefore \begin{cases} A \cap \bar{B} \cap A = B \cap \bar{A} \cap A & \text{即 } A \cap \bar{B} = \phi. \therefore A \subseteq B \\ A \cap \bar{B} \cap B = B \cap \bar{A} \cap B & \text{即 } \phi = B \cap \bar{A}. \therefore B \subseteq A \end{cases}$$

$$\therefore A=B$$

$$\text{充分性: } \because A=B. \therefore A-B=B-A=\phi$$

3.4

$$3. U = \{x \in \mathbb{Z} \mid 1 < x < 200\}$$

$$A_1 = \{x \in U \mid x \text{ 能被 } 2 \text{ 整除}\}$$

$$A_2 = \{x \in U \mid x \text{ 能被 } 3 \text{ 整除}\}$$

$$A_3 = \{x \in U \mid x \text{ 能被 } 5 \text{ 整除}\}$$

$$A_4 = \{x \in U \mid x \text{ 能被 } 7 \text{ 整除}\}$$

$$|A_1| = \lfloor \frac{200-1}{2} \rfloor = 99, |A_2| = \lfloor \frac{200-1}{3} \rfloor = 66$$

$$|A_3| = 39, |A_4| = 28, |A_1 \cap A_2| = 33, |A_1 \cap A_3| = 19$$

$$|A_1 \cap A_4| = 14, |A_2 \cap A_3| = 13, |A_2 \cap A_4| = 9$$

$$|A_3 \cap A_4| = 5, |A_1 \cap A_2 \cap A_3| = 6, |A_1 \cap A_2 \cap A_4| = 4$$

$$|A_1 \cap A_3 \cap A_4| = 2, |A_2 \cap A_3 \cap A_4| = 1, |A_1 \cap A_2 \cap A_3 \cap A_4| = 0$$

$$\therefore |A_1 \cup A_2 \cup A_3 \cup A_4| = 99 + 66 + 39 + 28 - 33 - 19 - 14 - 13 - 9 - 5 + 6 + 4 + 2 + 1 - 0 = 152 \text{ (若含 } 200 \text{ 则答案为 } 153)$$

$$4. J = \{\text{学日语的学生}\} \quad F = \{\text{学法语的学生}\}$$

$$E = \{\text{学英语的学生}\}, U = \{\text{所有学生}\}$$

$$|J| = 32, |F| = 20, |E| = 45, |U| = 100$$

$$|J \cup E| = 15, |J \cup F| = 7, |F \cup E| = 10, |\bar{E} \cap \bar{F} \cap J| = 30$$

$$(1) |E \cup F \cup J| = |U| - |\bar{E} \cap \bar{F} \cap J| = 70$$

$$= 32 + 20 + 45 - (15 + 7 + 10) + |E \cap F \cap J|$$

$$\text{得 } |E \cap F \cap J| = 5$$

$$(2) \text{学英语: } |E| - |E \cap F| - |E \cap J| + |J \cap E \cap F|$$

$$= 15$$

$$\text{学法语: } |F| - |F \cap E| - |F \cap J| + |F \cap E \cap J|$$

$$= 8$$

$$\text{学英语: } |E| - |E \cap F| - |E \cap J| + |J \cap E \cap F|$$

$$= 25$$

$$(13) 100 - 30 - 15 - 8 - 25 = 22$$

3.5

$$2. \text{否定: } A = \{1, 0\} \quad B = \{0\} \quad C = \{1\}$$

$$A \times A = \{\langle 1, 1 \rangle, \langle 1, 0 \rangle, \langle 0, 1 \rangle, \langle 0, 0 \rangle\}$$

$$B \times C = \{\langle 0, 1 \rangle\}$$

$$(A-B) \times (A-C) = \{\langle 1, 0 \rangle\}$$

$$\text{左边} = \{\langle 1, 1 \rangle, \langle 1, 0 \rangle, \langle 0, 0 \rangle\} \neq \text{右边}$$

$$5. \text{不一定. } A = \phi \text{ 时 } B \subseteq C \text{ 不恒成立}$$

$$A \neq \phi \text{ 时, } \forall \langle a, x \rangle \in A \times B, \exists \langle a, x \rangle \in A \times C.$$

$$\therefore \forall x \in B, x \in C. \text{ 故 } B \subseteq C$$

$$8. (1) \text{否定: } A = \{1\}, B = \{0\}, C = \{1\}, D = \{0\}$$

$$(A \cup B) \times (C \cup D) = \{\langle 1, 1 \rangle, \langle 1, 0 \rangle, \langle 0, 1 \rangle, \langle 0, 0 \rangle\}$$

$$(A \times C) \cup (B \times D) = \{\langle 1, 1 \rangle, \langle 0, 0 \rangle\}$$

$$(2) \text{证明: } \forall \langle a, b \rangle \in (A \cap B) \times (C \cap D)$$

$$\Leftrightarrow a \in A \text{ 且 } a \in B, b \in C \text{ 且 } b \in D.$$

$$\Leftrightarrow \langle a, b \rangle \in A \times C \text{ 且 } \langle a, b \rangle \in B \times D$$

$$\Leftrightarrow \langle a, b \rangle \in (A \times C) \cap (B \times D)$$

$$(3) \text{否定. } A = \{1, 0\} \quad B = \{1, 2\} \quad C = \{2\}, D = \{0\}$$

$$\text{左边} = \{\langle 0, 2 \rangle\}$$

$$\text{右边} = \{\langle 1, 2 \rangle, \langle 0, 2 \rangle\}$$

$$(4) \text{否定: } A = \{1\}, B = \{2\}, C = \{3\}, D = \{4\}$$

$$\text{左边} = \{\langle 1, 3 \rangle, \langle 1, 4 \rangle, \langle 2, 3 \rangle, \langle 2, 4 \rangle\}$$

$$\text{右边} = \{\langle 1, 3 \rangle, \langle 2, 4 \rangle\}$$

$$(5) \text{证明: } \forall \langle a, b \rangle \in (A-B) \times C.$$

$$\Leftrightarrow \forall a \in A-B, \forall b \in C$$

$$\Leftrightarrow \forall a, a \in A \text{ 且 } a \notin B, \forall b, b \in C$$

$$\Leftrightarrow \forall \langle a, b \rangle \in A \times C \text{ 且 } \forall \langle a, b \rangle \notin B \times C$$

$$\Leftrightarrow \forall \langle a, b \rangle \in A \times C - B \times C$$